SHIYU ZHOU

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Education

- **City University of Hong Kong, Hong Kong SAR** — Sept. 2022 – Sept. 2026 (expected)
  Ph.D. Candidate | GPA: 4.02/4.3 — *Supervisor: Prof. Gang Feng (IEEE Fellow)*
  Co-supervisor: Prof. Dong Sun (IEEE Fellow)
- **Nanyang Technological University** — Feb. 2026 – Aug. 2026
  Visiting Ph.D. Student — *Supervisor: Prof. Lihua Xie (IEEE Fellow)*
- **Beihang University, Beijing, China** — Sept. 2019 – July 2022
  M.Eng. in Control Science and Engineering | GPA: 91.97/100 | Rank: 1/30 — *Supervisor: Prof. Zhang Ren*
  Co-supervisor: Prof. Xiwang Dong
- **Northwestern Polytechnical University, Xi'an, China** — Sept. 2015 - July 2019
  B.Eng. in Automation | GPA: 92.38/100 | Rank: 1/118

Research Interests

- Cooperative control of multi-agent systems with applications to robotic manipulators, aerial robots, and ground robots.
- Fixed-time convergence algorithms for complex dynamical systems and non-cooperative games.
- Safety-critical control with collision avoidance and resilient control under communication failures and cyberattacks.

Publications

Selected Journal Papers

1. Shiyu Zhou, Dong Sun, Gang Feng*, "Distributed adaptive fixed-time formation tracking for heterogeneous multi-agent systems", ***Automatica***, vol. 183, no. 112632, 2026. [PDF][DOI]
2. Shiyu Zhou, Dong Sun, Xiwang Dong, Gang Feng*, "Fixed-time formation-containment tracking of heterogeneous multi-agent systems", ***IEEE Transactions on Automation Science and Engineering***, Early Access, 2026. [DOI]
3. Shiyu Zhou, Dong Sun, Gang Feng*, "Fixed-time formation tracking for heterogeneous linear multi-agent systems with a nonautonomous leader", ***IEEE Transactions on Control of Network Systems***, vol. 12, no. 2, pp. 1733-1743, 2025. [PDF][DOI]
4. Shiyu Zhou, Xiwang Dong*, Yongzhao Hua, Jianglong Yu, Zhang Ren, "Predefined formation-containment control of high-order multi-agent systems under communication delays and switching topologies", ***IET Control Theory and Applications***, vol. 12, no. 15, pp. 1661-1672, 2021. [PDF][DOI]
5. Shiyu Zhou, Dong Sun, Gang Feng*, "Fixed-time time-varying group formation tracking of heterogeneous linear multi-agent systems", ***Unmanned Systems***, vol. 14, no. 3, pp. 561–571, 2026.
6. Shiyu Zhou, Yongzhao Hua*, Xiwang Dong, Jianglong Yu, Zhang Ren, "Time-varying output formation-tracking of heterogeneous multi-agent systems with time-varying delays and switching topologies", ***Measurement and Control***, vol. 54, no. 9, pp. 1371-1382, 2021.
7. Shiyu Zhou*, Dong Sun, "Group formation tracking for heterogeneous linear multi-agent systems under switching topologies", ***Journal of Automation and Intelligence***, vol. 4, no. 2, pp. 108-114, 2025.
8. Shiyu Zhou, Yongzhao Hua, Xiwang Dong*, Qingdong Li, Zhang Ren, "Predefined containment control for general linear multi-agent systems with time-varying delays and switching topologies", ***Advanced Control for Applications: Engineering and Industrial Systems***, vol. 2, no. 2, pp. 1-12, 2019.
9. Weihao Li, Shiyu Zhou, Mengji Shi*, Jiangfeng Yue, Boxian Lin, Kaiyu Qin, "Collision avoidance time-varying group formation tracking control for multi-agent systems", ***Applied Intelligence***, vol. 55, no. 175, pp. 1-12, 2025.
10. Shiyu Zhou, Lihua Xie, Xiwang Dong, Dong Sun, Gang Feng*, "Collision-free practical fixed-time formation tracking for uncertain Euler-Lagrange systems", ***IEEE Transactions on Cybernetics***, 2026, under review.
11. Zhipeng Shen, Jinjie Li, Shiyu Zhou, Moju Zhao, Hailong Huang*, "Efficient trajectory optimization for generalized multirotors via sequential convex programming and convexity exploitation", ***IEEE Transactions on Automation Science and Engineering***, 2026, under second-round review.

Conference Papers

12. Shiyu Zhou*, Dong Sun, Gang Feng, "Time-varying output group formation tracking control for heterogeneous multi-agent systems with switching topologies", ***IEEE International Conference on Control and Automation***, 2024.
13. Shiyu Zhou, Zhipeng Shen, Jianglong Yu*, Xiwang Dong, Zhang Ren, "Output group formation-tracking control for heterogeneous systems with collision avoidance and connectivity maintenance", ***Chinese Control Conference***, 2022.
14. Shiyu Zhou, Xiwang Dong*, Qingke Tan, Qing Wang, Zhang Ren, "Time-varying group formation-tracking for general linear multi-agent systems with switching topologies and time-varying delays", ***IEEE International Conference on Industrial Technology***, 2021.

Research Projects

1. **Fixed-time cooperative control of multi-agent systems under denial-of-service attacks and its application**
   This project aims to explore and develop novel theories and methodologies for the fixed-time cooperative control problem of multi-agent systems under denial-of-service attacks, ensuring secure and reliable control in complex communication environments.
   - Funded by the ***Research Grants Council of Hong Kong*** (Grant No. CityU-11205024).
   - **Research Focus**
     - Adaptive fixed-time cooperative control of multi-agent systems under directed communication graphs.
     - Fixed-time cooperative control of multi-agent systems under denial-of-service attacks over directed communication graphs.
     - Fixed-time cooperative control of uncertain Euler–Lagrange systems, with potential applications to networked robotic manipulators.
2. **Theory and application for cooperative control of heterogeneous swarm systems in complex environments**
   This project is designed to address high-security requirements for large-scale heterogeneous swarm systems operating in complex and communication-denied environments, improving their robustness and adaptability.
   - Funded by the ***Science and Technology Innovation 2030 - Key Project of "New Generation Artificial Intelligence"*** (Grant No. 2020AAA0108200).
   - **Research Focus**
     - Cooperative control of heterogeneous multi-agent systems under time-varying delays and switching topologies.
     - Robust cooperative control under parameter uncertainties and external disturbances.
     - Collision and obstacle avoidance for heterogeneous multi-agent systems.

Honors and Awards

Scholarships

- **China National Scholarship (Graduate)**, Ministry of Education — 2021
- **China National Scholarship (Undergraduate)**, Ministry of Education — 2016, 2017, 2018
- Research Tuition Scholarship, City University of Hong Kong — 2025
- First Prize Scholarship, Beihang University — 2020, 2021
- First Prize Scholarship, Northwestern Polytechnical University — 2016, 2017, 2018

Awards

- Outstanding Academic Performance Award for Research Degree Students, City University of Hong Kong — 2025
- Outstanding Open Science Author of the Year 2022, Wiley — 2022
- Finalist for Excellent Dissertation Award, Chinese Society of Aeronautics and Astronautics — 2022
- Excellent Master Dissertation Award, Beihang University — 2022
- Excellent Bachelor's Dissertation Award, Northwestern Polytechnical University — 2019
- Honorable Mention, Interdisciplinary Contest in Modeling — 2017
- First Prize, China Undergraduate Mathematical Contest in Modeling — 2017
- First Prize, Northwestern Polytechnical University Eighth Smart Car Competition — 2018
- First Prize, May Day Mathematical Contest in Modeling — 2017, 2018
- Second Prize, National University Students Electrical Math Modeling Competition — 2017
- Second Prize, Social Practice and Science Contest on Energy Saving and Emission Reduction — 2017
- Third Prize, China Post-Graduate Mathematic Contest in Modeling — 2020
- Third Prize, Chinese Mathematics Competitions (Non-Mathematics Major) — 2017

Honors

- Outstanding Graduate Student of Beijing, Beijing Municipal Education Commission — 2022
- Outstanding Graduate Student, Beijing Association of Automation — 2022
- Top Ten Outstanding Graduate Students, Beihang University — 2021
- Outstanding Undergraduate Student, Northwestern Polytechnical University — 2019
- Merit Student, Beihang University — 2021, 2022
- Merit Student, Northwestern Polytechnical University — 2016, 2017, 2018
- Top Five Runner, City University of Hong Kong — 2026, 2025

Skills and Interests

- **Language:** Chinese (Native), English (Fluent, TOEFL:94, CET-6:554)
- **Programming:** MATLAB, Python, C/C++
- **Interests:** Marathon (Top 5 Runner of CityUHK), Badminton, Hiking, Photography

Professional Services

- Reviewer: IEEE Transactions on Automatic Control, Systems and Control Letters, IEEE/CAA Journal of Automatica Sinica
- Memberships: IEEE Student Member, IEEE Control and Systems Society Member